16. Memory Protection Unit (MPU)

16.1 Overview

EDMAC in this chapter refers to the GWCA function of ESWM.

All bus masters have Memory Protection Units (MPUs).

Table 16.1 lists the MPU specifications, and Table 16.2 shows the behavior on detection of each MPU error.

Table 16.1 MPU specifications

Classification	Module/Function	Specifications
Memory protection	Arm MPU	Memory protection function for the CPU: ● CPU0: Secure MPU 8 regions and Non-secure MPU 8 regions ● CPU1: Secure MPU 8 regions and Non-secure MPU 8 regions
	Bus master MPU	Memory protection function for each bus master except for the CPU: ● NPU: 5 regions ● DMAC0 (DMAC/DTC0): 8 regions ● DMAC1 (DMAC/DTC1): 8 regions ● EDMAC (Ether-DMAC): 5 regions ● GLCDC: 2 regions ● DRW: 3 regions ● MIPI-DSI: 1 region ● MIPI-CSI by VIN: 3 regions ● CEU: 2 regions

Table 16.2 Behavior on MPU error detection

MPU type	Access permissions setting	Boundary address setting minimum unit	Error response for the MPU error notification	Bus access at error detection	Hold the information of error access
Arm MPU	Read access Write access Execution	32 bytes	Supported*1	● Incorrectly write access ● Incorrectly read access	Hold in CPU
Bus master MPU	Read access Write access Privileged access (DMAC/DTC only)	NPU: 4 KB DMAC0/1: 32 bytes EDMAC: 32 bytes GLCDC: 1 KB DRW: 1 KB MIPI-DSI: 4 KB MIPI-CSI: 4 KB CEU: 4 KB	Supported	● Write access ignored ● Read access is read as 0	Hold

Note 1. A privileged DAP request through the unprivileged debug extension mechanism is demoted to an unprivileged access and is subject to MPU checks. Both privileged and unprivileged requests are subject to MPU checks.

For information on error access for the Arm MPU, see section 16.4. References. For information on error access for other MPUs, see section 16.3.1. Register Descriptions and section 15.7. Bus Error Monitoring Section.

16.2 Arm MPU

The Arm MPU provides full support for:

- CPU0, 1: 8 secure regions and 8 non-secure regions
- Access permissions
- Export of memory attributes to the system

Arm MPU mismatches and permission violations invoke the programmable-priority MemManage fault (Hard Fault) handler. For details, see section 16.4. References.

16.3 Bus Master MPU

The bus master MPU monitors the addresses accessed by the bus masters in the entire address space (0x0000_0000 to 0xFFFF_FFFF). The access control information can be set up to:

- 5 regions in NPU
- 8 regions in DMAC0
- 8 regions in DMAC1
- 5 regions in EDMAC
- 2 regions in GLCDC
- 3 regions in DRW
- 1 region in MIPID
- 3 regions in MIPIC
- 2 regions in CEU

The bus master MPU monitors access to each region based on these settings.

If accesses violate the access permissions that are configured in the bus master MPU or crossing the boundaries of a valid MPU region with burst access, the bus master MPU returns an error response. For information on error access, see section 16.3.1. Register Descriptions and section 15.7.2. Operations When a Bus Error Occurs.

Table 16.3 lists the specifications of the bus master MPU.

Table 16.3 Bus master MPU specifications

Parameter	Description
Master groups	(Group name: Corresponding bus master) ● NPU: NPU0/NPU1 ● DMAC0: DMAC/DTC0 ● DMAC1: DMAC/DTC1 ● EDMAC: Ether-DMAC ● GLCDC: GLCDC0/GLCDC1 ● DRW: DRW0/DRW1 ● MIPID: MIPI-DSI ● MIPIC: MIPI-CSI by VIN ● CEU: CEU
Protected regions	0x0000_0000 to 0xFFFF_FFFF
Number of regions	● NPU: 5 regions ● DMAC0: 8 regions ● DMAC1: 8 regions ● EDMAC: 5 regions ● GLCDC: 2 regions ● DRW: 3 regions ● MIPID: 1 region ● MIPIC: 3 regions ● CEU: 2 regions
Address specification for individual regions	● Region start and end addresses are configurable
Enable or disable setting for memory protection in individual regions	● Settings enabled or disabled for the associated region
Access-control settings for individual regions	● Permission to read and write ● Permission to unprivileged access
Operation on error detection	● Reset or error response
Register protection	● Register can be protected from illegal writes

16.3.1 Register Descriptions

Register writes processing should be performed after stopping the corresponding master group bus access.

SA (Secure Attribution) is set in each register and the SA is used to determine whether the target register is a secure register or a non-secure register.

16.3.1.1 MMPUSARA : Master Memory Protection Unit Security Attribution Register A

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x130

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	MMPUASAn[23:16]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	MMPUASAn[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	MMPUASA7 to MMPUASA0	MMPUAn Security Attribution (n = 0 to 7) 0: Secure 1: Non-secure	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	MMPUASA23 to MMPUASA16	MMPUAn Security Attribution (n = 16 to 23) 0: Secure 1: Non-secure	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1
P-TYPE-1

MMPUASA bits (MMPUAn Security Attribution (n = 0 to 7))

These bits specify security attributes of the Bus Master MPU Region Setting register for DMAC0. The target registers are as follows:

- MMPUSDMAC00n
- MMPUEDMAC00n
- MMPUACDMAC00n

MMPUASA bits (MMPUAn Security Attribution (n = 16 to 23))

These bits specify security attributes of the Bus Master MPU Region Setting register for DMAC1. The target registers are as follows:

- MMPUSDMAC10y (y = n − 16)
- MMPUEDMAC10y (y = n − 16)
- MMPUACDMAC10y (y = n − 16)

16.3.1.2 MMPUSARB : Master Memory Protection Unit Security Attribution Register B

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x134

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MMPU BSA8	—	—	—	—	—	—	MMPU BSA1	MMPU BSA0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MMPUBSA0	MMPUB0 Security Attribution 0: Secure 1: Non-secure	R/W
1	MMPUBSA1	MMPUB1 Security Attribution 0: Secure 1: Non-secure	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	MMPUBSA8	MMPUB8 Security Attribution 0: Secure 1: Non-secure	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1
P-TYPE-1

MMPUBSA0 bit (MMPUB0 Security Attribution)

This bit specifies the security attributes of the DMAC0 MPU Enable Setting register. The target registers are as follows:

- MMPUENDMAC0
- MMPUENPTDMAC0

MMPUBSA1 bit (MMPUB1 Security Attribution)

This bit specifies the security attributes of the DMAC1 MPU Enable Setting register. The target registers are as follows:

- MMPUENDMAC1
- MMPUENPTDMAC1

MMPUBSA8 bit (MMPUB8 Security Attribution)

This bit specifies the security attributes of the MPU Operation After Detection Setting register. The target registers are as follows:

- MMPUOAD
- MMPUOADPT

16.3.1.3 MMPUSDMACmn : MMPU Start Address Register for DMAC (m = 0, 1, n = 00 to 07)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0204 + 0x0010 × n (DMAC0n)
0x0404 + 0x0010 × n (DMAC1n)

Bit position: 31 … 0

Bit field: (blank)

Value after reset: x x x x x x x x x x x x x x x x x x x x x x x x x x x 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_FFE0. Writing is ignored for bit 0 to bit 4. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSDMACmn (m = 0, 1, n = 00 to 07) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

Regions set by MMPUSDMACmn (m = 0, 1, n = 00 to 07), MMPUEDMACmn (m = 0, 1, n = 00 to 07) and MMPUACDMACmn (m = 0, 1, n = 00 to 07) registers, can be set for a secure access or a non-secure access with the MMPUSARA register. If the corresponding MMPUSARA.MMPUASAn (n = 0 to 7, 16 to 23) bit is set to 1, it is only possible to use that region with a non-secure access. On the other hand, if the corresponding MMPUSARA.MMPUASAn (n = 0 to 7, 16 to 23) bit is set to 0, it is only possible to use that region with a secure access.

16.3.1.4 MMPUSEDMACn : MMPU Start Address Register for EDMAC (n = 0 to 4)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0604 + 0x010 × n

Bit position: 31 … 0

Bit field: (blank)

Value after reset: x x x x x x x x x x x x x x x x x x x x x x x x x x x 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for EDMAC Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_FFE0. Writing is ignored for bit 0 to bit 4. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSEDMACn (n = 0 to 4) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.5 MMPUSGLCDCn : MMPU Start Address Register for GLCDC (n = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0804 + 0x010 × n

Bit position: 31 ... 0

Bit field:

Value after reset: x x x x x x x x x x x x x x x x x x x x x x 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for GLCDC Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_FC00. Writing is ignored for bit 0 to bit 9. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSGLCDCn (n = 0, 1) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.6 MMPUSDRWn : MMPU Start Address Register for DRW (n = 0 to 2)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0A04 + 0x010 × n

Bit position: 31 ... 0

Bit field:

Value after reset: x x x x x x x x x x x x x x x x x x x x x x 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for DRW Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_FC00. Writing is ignored for bit 0 to bit 9. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSDRWn (n = 0 to 2) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.7 MMPUSMIPID : MMPU Start Address Register for MIPI-DSI

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0C04

Bit position: 31 ... 0

Bit field:

Value after reset: x x x x x x x x x x x x x x x x x x x x 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for MIPI-DSI Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_F000. Writing is ignored for bit 0 to bit 11. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSMIPID register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.8 MMPUSCEUn : MMPU Start Address Register for CEU (n = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0E04+ 0x010 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for CEU Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_F000. Writing is ignored for bit 0 to bit 11. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSCEUn (n = 0, 1) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.9 MMPUSMIPICn : MMPU Start Address Register for MIPI-CSI via VIN (n = 0 to 2)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1004 + 0x0010 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for MIPI-CSI. Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_F000. Writing is ignored for bit 0 to bit 11. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSMIPICn (n = 0 to 2) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.10 MMPUSNPUn : MMPU Start Address Register for NPU (n = 0 to 4)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1204 + 0x0010 × n

Bit position:	31																														0
Bit field:																															
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	n/a	Region start address register for NPU. Address where the region starts, for use in region determination. The starting address of the MPU area must be set in the range of 0x0000_0000 to 0xFFFF_F000. Writing is ignored for bit 0 to bit 11. These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUSNPUn (n = 0 to 4) register specifies the start address where the region starts.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.11 MMPUEDMACmn : MMPU End Address Register for DMAC (m = 0, 1, n = 00 to 07)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0208 + 0x0010 × n (DMAC0n)
0x0408 + 0x0010 × n (DMAC1n)

Bit position:	31																														0
Bit field:																															
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1	1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register Address where the region end, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_001F to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 4. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUEDMACmn (m = 0, 1, n = 00 to 07) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.12 MMPUEEDMACn : MMPU End Address Register for EDMAC (n = 0 to 4)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0608 + 0x0010 × n

Bit position:	31 … 0
Bit field:	
Value after reset:	x x x x x x x x x x x x x x x x x x x x x x x x x x x 1 1 1 1 1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for EDMAC Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_001F to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 4. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUEEDMACn (n = 0 to 4) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.13 MMPUEGLCDCn : MMPU End Address Register for GLCDC (n = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0808 + 0x010 × n

Bit position:	31 … 0
Bit field:	
Value after reset:	x x x x x x x x x x x x x x x x x x x x x x 1 1 1 1 1 1 1 1 1 1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for GLCDC Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_03FF to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 9. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUEGLCDCn (n = 0, 1) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.14 MMPUEDRWn : MMPU End Address Register for DRW (n = 0 to 2)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0A08 + 0x010 × n

Bit position:	31 … 0
Bit field:	
Value after reset:	x x x x x x x x x x x x x x x x x x x x x x 1 1 1 1 1 1 1 1 1 1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for DRW Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_03FF to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 9. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUEDRWn (n = 0 to 2) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.15 MMPUEMIPID : MMPU End Address Register for MIPI-DSI

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0C08

Bit position:	31																														0
Bit field:																															
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for MIPI-DSI Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_0FFF to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 11. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUEMIPID register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.16 MMPUECEUn : MMPU End Address Register for CEU (n = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0E08 + 0x010 × n

Bit position:	31																														0
Bit field:																															
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for CEU Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_0FFF to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 11. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUECEUn (n = 0, 1) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.17 MMPUEMIPICn : MMPU End Address Register for MIPI-CSI via VIN (n = 0 to 2)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1008 + 0x0010 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for MIPI-CSI Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_0FFF to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 11. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUEMIPICn (n = 0 to 2) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.18 MMPUENPUn : MMPU End Address Register for NPU (n = 0 to 4)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1208 + 0x0010 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:																																
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
31:0	n/a	Region end address register for NPU Address where the region ends, for use in region determination. The ending address of the MPU area must be set in the range of 0x0000_0FFF to 0xFFFF_FFFF. Writing is ignored for bit 0 to bit 11. These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3
P-TYPE-2

The MMPUENPUn (n = 0 to 4) register specifies the end address where the region ends.

This register requires word access. Byte access and halfword access are prohibited. When byte access and halfword access are executed, operation is not guaranteed.

16.3.1.19 MMPUACDMACmn : MMPU Access Control Register for DMAC (m = 0, 1, n = 00 to 07)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0200 + 0x0010 × n (DMAC0n)
0x0400 + 0x0010 × n (DMAC1n)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PP	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: DMAC region n unit is disabled 1: DMAC region n unit is enabled	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
3	PP	Privilege protection 0: Unprivileged access permission 1: Unprivileged access protection	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the DMACm region n (m = 0, 1, n = 00 to 07) unit.

When the ENABLE bit is set to 1, the RP bit, the WP bit and the PP bit are enabled for permit or protect access to the region that is set in MMPUSDMACmn (m = 0, 1, n = 00 to 07) and MMPUEDMACmn (m = 0, 1, n = 00 to 07).

When the ENABLE bit is set to 0, the access to DMACm region n (m = 0, 1, n = 00 to 07) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for DMACm region n (m = 0, 1, n = 00 to 07).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for DMACm region n (m = 0, 1, n = 00 to 07).

When the ENABLE bit is set to 1, the WP bit is available.

PP bit (Privilege protection)

The PP bit enables or disables unprivileged access protection for DMACm region n (m = 0, 1, n = 00 to 07).

When the ENABLE bit is set to 1, the PP bit is available.

16.3.1.20 MMPUACEDMACn : MMPU Access Control Register for EDMAC (n = 0 to 4)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0600 + 0x0010 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: EDMAC region n unit is disabled. 1: EDMAC region n unit is enabled.	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the EDMAC region n (n = 0 to 4) unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSEDMACn (n = 0 to 4) and MMPUEEDMACn (n = 0 to 4).

When the ENABLE bit is set to 0, the access to EDMAC region n (n = 0 to 4) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for EDMAC region n (n = 0 to 4).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for EDMAC region n (n = 0 to 4).

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.21 MMPUACGLCDCn : MMPU Access Control Register for GLCDC (n = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0800 + 0x010 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: GLCDC region n unit is disabled 1: GLCDC region n unit is enabled	R/W

Bit	Symbol	Function	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the GLCDC region n (n = 0, 1) unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSGLCDCn (n = 0, 1) and MMPUEGLCDCn (n = 0, 1).

When the ENABLE bit is set to 0, the access to GLCDC region n (n = 0, 1) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for GLCDC region n (n = 0, 1).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for GLCDC region n (n = 0, 1).

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.22 MMPUACDRWn : MMPU Access Control Register for DRW (n = 0 to 2)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0A00 + 0x010 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: DRW region n unit is disabled 1: DRW region n unit is enabled	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the DRW region n (n = 0 to 2) unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSDRWn (n = 0 to 2) and MMPUEDRWn (n = 0 to 2).

When the ENABLE bit is set to 0, the access to DRW region n (n = 0 to 2) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for DRW region n (n = 0 to 2).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for DRW region n (n = 0 to 2).

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.23 MMPUACMIPID : MMPU Access Control Register for MIPI-DSI

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0C00

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: MIPI DSI region unit is disabled 1: MIPI DSI region unit is enabled	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the MIPI-DSI region unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSMIPID and MMPUEMIPID.

When the ENABLE bit is set to 0, the access to MIPI-DSI region is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for MIPI-DSI region.

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for MIPI-DSI region.

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.24 MMPUACCEUn : MMPU Access Control Register for CEU (n = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0E00 + 0x010 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: CEU region n unit is disabled 1: CEU region n unit is enabled	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the CEU region n (n = 0, 1) unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSCEUn (n = 0, 1) and MMPUECEUn (n = 0, 1).

When the ENABLE bit is set to 0, the access to CEU region n (n = 0, 1) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for CEU region n (n = 0, 1).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for CEU region n (n = 0, 1).

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.25 MMPUACMIPICn : MMPU Access Control Register for MIPI-CSI via VIN (n = 0 to 2)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1000 + 0x0010 × n

Bit position:	15													2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: MIPI region n unit is disabled 1: MIPI region n unit is enabled	R/W

Bit	Symbol	Function	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the MIPI-CSI region n (n = 0 to 2) unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSMIPICn (n = 0 to 2) and MMPUEMIPICn (n = 0 to 2).

When the ENABLE bit is set to 0, the access to MIPI-CSI region n (n = 0 to 2) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for MIPI-CSI region n (n = 0 to 2).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for MIPI-CSI region n (n = 0 to 2).

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.26 MMPUACNPUn : MMPU Access Control Register for NPU (n = 0 to 4)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1200 + 0x0010 × n

Bit position:	15													2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	WP	RP	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Region Enable 0: NPU region n unit is disabled 1: NPU region n unit is enabled	R/W
1	RP	Read protection 0: Read permission 1: Read protection	R/W
2	WP	Write protection 0: Write permission 1: Write protection	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

ENABLE bit (Region Enable)

The ENABLE bit enables or disables the NPU region n (n = 0 to 4) unit.

When the ENABLE bit is set to 1, the RP bit and the WP bit are enabled for permit or protect access to the region that is set in MMPUSNPUn (n = 0 to 4) and MMPUENPUn (n = 0 to 4).

When the ENABLE bit is set to 0, the access to NPU region n (n = 0 to 4) is not specified.

RP bit (Read protection)

The RP bit enables or disables read protection for NPU region n (n = 0 to 4).

When the ENABLE bit is set to 1, the RP bit is available.

WP bit (Write protection)

The WP bit enables or disables write protection for NPU region n (n = 0 to 4).

When the ENABLE bit is set to 1, the WP bit is available.

16.3.1.27 MMPUENDMACm : MMPU Enable Register for DMAC (m = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0100 (DMAC0n)
0x0300 (DMAC1n)

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of DMAC Enable 0: Bus master MPU of DMAC is disabled 1: Bus master MPU of DMAC is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of DMAC Enable)

The ENABLE bit enables or disables the bus master MPU function for DMACm (m = 0, 1).

When the ENABLE bit is set to 1, MMPUACDMACmn (m = 0, 1, n = 00 to 07) is available.

When the ENABLE bit is set to 0, MMPUACDMACmn (m = 0, 1, n = 00 to 07) is unavailable including permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.28 MMPUENEDMAC : MMPU Enable Register for EDMAC

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0500

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of EDMAC Enable 0: Bus master MPU of EDMAC is disabled 1: Bus master MPU of EDMAC is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of EDMAC Enable)

The ENABLE bit enables or disables the bus master MPU function of each master group.

When the ENABLE bit is set to 1, MMPUACEDMACn (n = 0 to 3) is available.

When the ENABLE bit is set to 0, MMPUACEDMACn (n = 0 to 3) is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.29 MMPUENGLCDC : MMPU Enable Register for GLCDC

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0700

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of GLCDC Enable 0: Bus master MPU of GLCDC is disabled 1: Bus master MPU of GLCDC is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of GLCDC Enable)

The ENABLE bit enables or disables the bus master MPU function of each master group.

When the ENABLE bit is set to 1, MMPUACGLCDCn (n = 0, 1) is available.

When the ENABLE bit is set to 0, MMPUACGLCDCn (n = 0, 1) is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.30 MMPUENDRW : MMPU Enable Register for DRW

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0900

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of DRW Enable 0: Bus master MPU of DRW is disabled 1: Bus master MPU of DRW is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of DRW Enable)

The ENABLE bit enables or disables the bus master MPU function of each master group.

When the ENABLE bit is set to 1, MMPUACDRWn (n = 0 to 2) is available.

When the ENABLE bit is set to 0, MMPUACDRWn (n = 0 to 2) is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.31 MMPUENMIPID : MMPU Enable Register for MIPI-DSI

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0B00

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of MIPI-DSI Enable 0: Bus master MPU of MIPI-DSI is disabled 1: Bus master MPU of MIPI-DSI is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of MIPI-DSI Enable)

The ENABLE bit enables or disables the bus master MPU function of each master group.

When the ENABLE bit is set to 1, MMPUACMIPID is available.

When the ENABLE bit is set to 0, MMPUACMIPID is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.32 MMPUENCEU : MMPU Enable Register for CEU

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0D00

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of CEU Enable 0: Bus master MPU of CEU is disabled 1: Bus master MPU of CEU is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of CEU Enable)

The ENABLE bit enables or disables the bus master MPU function of each master group.

When the ENABLE bit is set to 1, MMPUACCEUn (n = 0, 1) is available.

When the ENABLE bit is set to 0, MMPUACCEUn (n = 0, 1) is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.33 MMPUENMIPIC : MMPU Enable Register for MIPI-CSI by VIN

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0F00

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of MIPI-CSI Enable 0: Bus master MPU of MIPI-CSI is disabled 1: Bus master MPU of MIPI-CSI is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of MIPI-CSI Enable)

The ENABLE bit enables or disables the bus master MPU-CSI function of each master group.

When the ENABLE bit is set to 1, MMPUACMIPICn (n = 0 to 2) is available.

When the ENABLE bit is set to 0, MMPUACMIPICn (n = 0 to 2) is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.34 MMPUENNPU : MMPU Enable Register for NPU

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1100

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	ENABLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENABLE	Bus master MPU of NPU Enable 0: Bus master MPU of NPU is disabled 1: Bus master MPU of NPU is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the ENABLE bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

ENABLE bit (Bus master MPU of NPU Enable)

The ENABLE bit enables or disables the bus master MPU function of each master group.

When the ENABLE bit is set to 1, MMPUACNPUn (n = 0 to 4) is available.

When the ENABLE bit is set to 0, MMPUACNPUn (n = 0 to 4) is unavailable, and permission for all regions.

The bus master MPU function of each master group sets the ENABLE bit.

When the ENABLE bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the ENABLE bit.

When writing the ENABLE bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.35 MMPUENPTDMACm : MMPU Enable Protect Register for DMAC m (m = 0, 1)

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0104 (DMAC0n)
0x0304 (DMAC1n)

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENDMACm register write is possible. 1: MMPUENDMACm register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENDMACm (m = 0, 1) register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.36 MMPUENPTEDMAC : MMPU Enable Protect Register for EDMAC

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0504

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENEDMAC register write is possible. 1: MMPUENEDMAC register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENEDMAC register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.37 MMPUENPTGLCDC : MMPU Enable Protect Register for GLCDC

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0704

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENGLCDC register write is possible. 1: MMPUENGLCDC register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENGLCDC register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.38 MMPUENPDRW : MMPU Enable Protect Register for DRW

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0904

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENDRW register write is possible. 1: MMPUENDRW register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENDRW register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.39 MMPUENPTMIPID : MMPU Enable Protect Register for MIPI-DSI

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0B04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENMIPID register write is possible. 1: MMPUENMIPID register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.

When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENMIPID register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.40 MMPUENPTCEU : MMPU Enable Protect Register for CEU

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0D04

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENCEU register write is possible. 1: MMPUENCEU register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENCEU register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.41 MMPUENPTMIPIC : MMPU Enable Protect Register for MIPI-CSI by VIN

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0F04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENMIPIC register write is possible. 1: MMPUENMIPIC register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENMIPIC register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.42 MMPUENPTNPU : MMPU Enable Protect Register for NPU

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1104

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUENNPU register write is possible. 1: MMPUENNPU register write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.

When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit controls protection of MMPUENNPU register.

When the PROTECT bit is set, write 0xA5 to the KEY[7:0] bits at the same time.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits are used to enable or disable writing to the PROTECT bit.

When writing the PROTECT bit, write 0xA5 to the KEY[7:0] bits at the same time.

When a value other than 0xA5 is written to the KEY[7:0] bits, the ENABLE bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.43 MMPURPTDMACm : MMPU Regions Protect Register for DMAC Non-secure m (m = 0, 1)

Base address: RMPU_NS = 0x5000_0000

Offset address: 0x0108 (DMAC0n)
0x0308 (DMAC1n)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for DMAC write is possible. 1: Bus master MPU register for DMAC write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-7
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTDMACm.PROTECT (m =0, 1) controls the following registers:

- MMPUSDMACmn (m =0, 1, n = 00 to 07) of non-secure
- MMPUEDMACmn (m =0, 1, n = 00 to 07) of non-secure
- MMPUACDMACmn (m =0, 1, n = 00 to 07) of non-secure

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.44 MMPURPTDMAC_SECm : MMPU Regions Protect Register for DMAC Secure m (m = 0, 1)

Base address: RMPU = 0x4000_0000

Offset address: 0x010C (DMAC0n)
0x030C (DMAC1n)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for DMAC Secure write is possible. 1: Bus master MPU register for DMAC Secure write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-6
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTDMAC_SECm.PROTECT (m = 0, 1) controls the following registers:

- MMPUSDMACmn (m = 0, 1, n = 00 to 07) of secure
- MMPUEDMACmn (m = 0, 1, n = 00 to 07) of secure
- MMPUACDMACmn (m = 0, 1, n = 00 to 07) of secure

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.45 MMPURPTEDMAC : MMPU Regions Protect Register for EDMAC

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0508

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for EDMAC write is possible. 1: Bus master MPU register for EDMAC write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTEDMAC.PROTECT controls the following registers:

- MMPUSEDMACn (n = 0 to 4)
- MMPUEEDMACn (n = 0 to 4)
- MMPUACEDMACn (n = 0 to 4)

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.46 MMPURPTGLCDC : MMPU Regions Protect Register for GLCDC

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0708

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for GLCDC write is possible. 1: Bus master MPU register for GLCDC write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTGLCDC.PROTECT controls the following registers:

- MMPUSGLCDCn (n = 0, 1)
- MMPUEGLCDCn (n = 0, 1)
- MMPUACGLCDCn (n = 0, 1)

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.47 MMPURPTDRW : MMPU Regions Protect Register for DRW

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0908

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for DRW write is possible. 1: Bus master MPU register for DRW write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPDRW.PROTECT controls the following registers:

- MMPUSDRWn (n = 0 to 2)
- MMPUEDRWn (n = 0 to 2)
- MMPUACDRWn (n = 0 to 2)

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.48 MMPURPTMIPID : MMPU Regions Protect Register for MIPI-DSI

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0B08

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for MIPI-DSI write is possible. 1: Bus master MPU register for MIPI-DSI write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTMIPID.PROTECT controls the following registers:

- MMPUSMIPID
- MMPUEMIPID
- MMPUACMIPID

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.49 MMPURPTCEU : MMPU Regions Protect Register for CEU

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0D08

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for CEU write is possible. 1: Bus master MPU register for CEU write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTCEU.PROTECT controls the following registers:

- MMPUSCEUn (n = 0, 1)
- MMPUECEUn (n = 0, 1)
- MMPUACCEUn (n = 0, 1)

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.50 MMPURPTMIPIC : MMPU Regions Protect Register for MIPI-CSI by VIN

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0F08

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for MIPI-CSI write is possible. 1: Bus master MPU register for MIPI-CSI write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTMIPIC.PROTECT controls the following registers:

- MMPUSMIPICn (n = 0 to 2)
- MMPUEMIPICn (n = 0 to 2)
- MMPUACMIPICn (n = 0 to 2)

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.51 MMPURPTNPU : MMPU Regions Protect Register for NPU

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x1108

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Bus master MPU register for NPU write is possible. 1: Bus master MPU register for NPU write is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPURPTNPU.PROTECT controls the following registers:

- MMPUSNPUn (n = 0 to 4)
- MMPUENPUn (n = 0 to 4)
- MMPUACNPUn (n = 0 to 4)

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When a value other than 0xA5 is written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.1.52 MMPUOAD : MMPU Operation After Detection Register

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0000

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	OAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OAD	Operation after detection 0: IRQ 1: Reset	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
15:8	KEY[7:0]	Key Code This bit is used to enable or disable writing to the OAD bit.	R/W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

OAD bit (Operation after detection)

The OAD bit specifies operation when the access violation is detected.

When OAD bit is 0, error response is returned to the bus master, and if BUSIRQEN<Master Name>.EN = 1, IRQ request is generated. For details, see section 15.3.32. BUSIRQEN<Master Name> : BUS Error Interrupt Enable Register . Interrupts due to bus errors can also be used as NMI. For details, see the specifications of the ICU.NMIER register.

When OAD bit is 1, reset request is generated.

When writing to the OAD bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the OAD bit. When writing to the OAD bit, write 0xA5 simultaneously to the KEY[7:0] bits. When other values are written, the OAD bit is not updated. The KEY[7:0] bits always read as 0x00.

16.3.1.53 MMPUOADPT : MMPU Operation After Detection Protect Register

Base address: RMPU = 0x4000_0000
RMPU_NS = 0x5000_0000

Offset address: 0x0004

Bit position:	15							8								0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: MMPUOAD register writing is possible. 1: MMPUOAD register writing is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits enable or disable writing to the PROTECT bit.	W

Note: S-TYPE-3
P-TYPE-2

Note: It is necessary to write by halfword access as byte-write access is prohibited.
When byte-write access is executed, operation is not guaranteed.

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writing to the associated registers to be protected.

MMPUOAD.PROTECT controls the following registers:

- MMPUOAD

When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits using halfword access.

KEY[7:0] bits (Key Code)

The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 simultaneously to the KEY[7:0] bits. When other values are written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

16.3.2 Operation

16.3.2.1 Memory Protection

The bus master MPU monitors memory access using control settings made individually for the access control regions. If accesses violate the access permissions that are configured in bus master MPU, the bus master MPU generates a memory protection error.

Bus master MPU has master groups of NPU, DMAC0, DMAC1, EDMAC, GLCDC, DRW, MIPI-DSI, MIPI-CSI and CEU. The memory protection function checks the address of the bus for a unified master group, and blocks illegal access of the master group to the protected region by bus master MPU.

The region setting registers of the bus master MPU for DMAC can be set for secure master and non-secure master using the MMPUSARA register. The MPU region setting with MMPUSARA.MMPUASAn = 0 applies only to access from the secure master, and the MPU region setting with MMPUSARA.MMPUASAn = 1 applies only to access from the non-secure master. Figure 16.1 shows an example of access authorization for the secure master and non-secure master when both secure and non-secure MPU settings are configured.

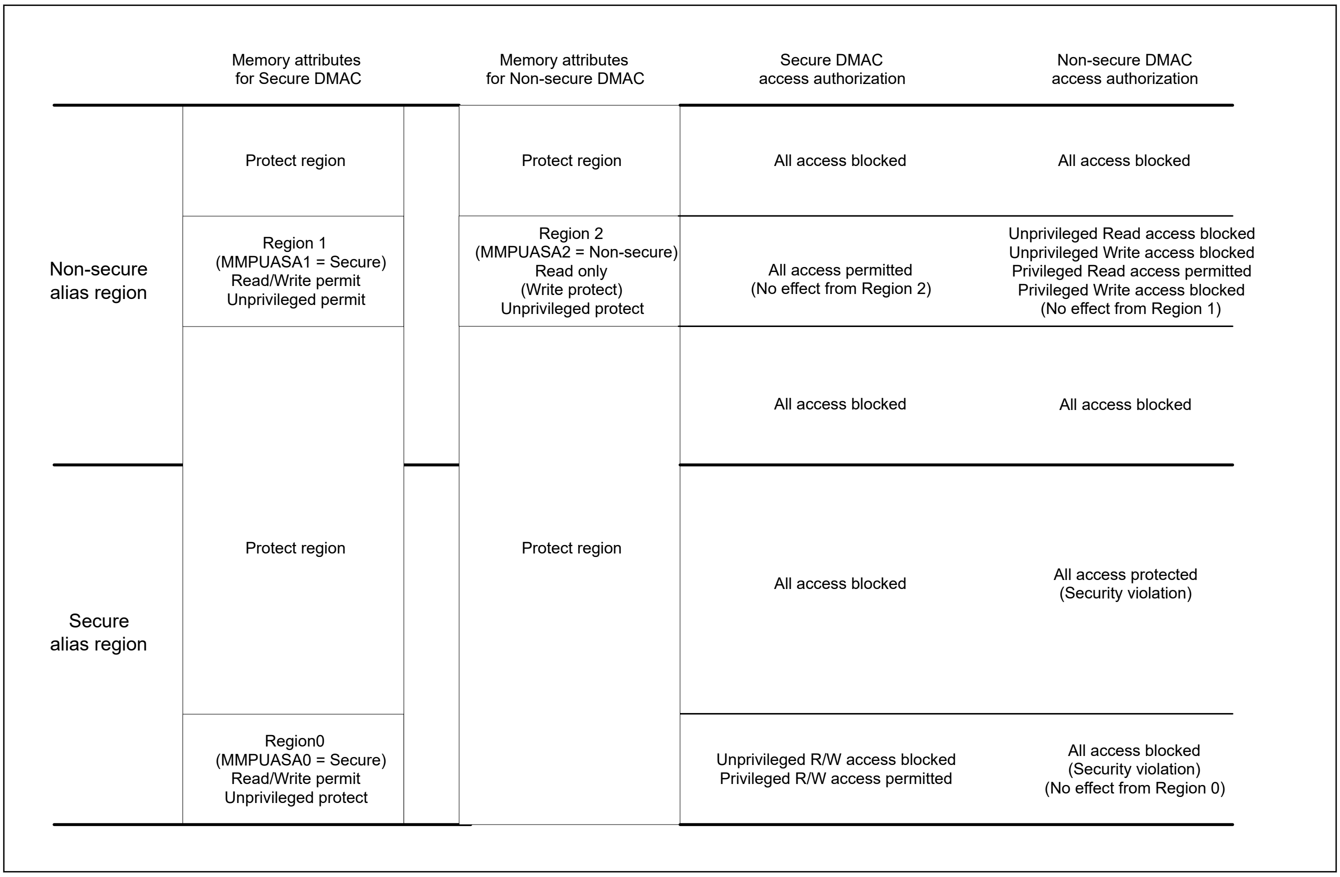

Figure 16.1 Example of access authority when both secure and non-secure regions of MPUs are set

Bus master MPU permit of all regions after reset. If MMPUENXXXX.ENABLE is 1 and there is no MPU region setting for the corresponding bus master, all regions are protected. (XXXX = Master Group name)

Figure 16.2 shows the access permission or protection by the overlapping bus master MPU regions.

Access control for the overlapping regions is as follows:

- The region is handled as a protected region when output of one or more region units is a protected region

- The region is handled as a protected region when output of all region units is outside of the regions
- Other cases are handled as permitted regions.

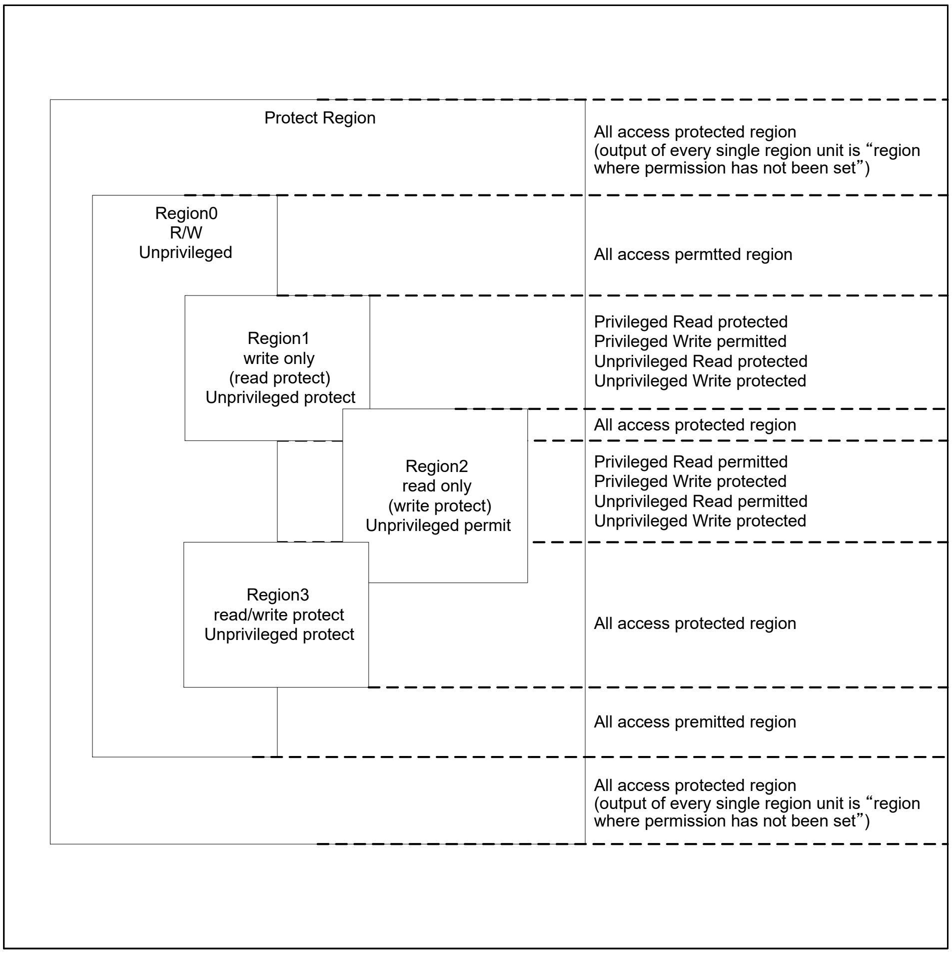

Figure 16.2 Access permission or protection by overlap of the bus master MPU regions

Figure 16.3 shows the register setting flow after reset. During this register setting, stop the bus master except the CPU.

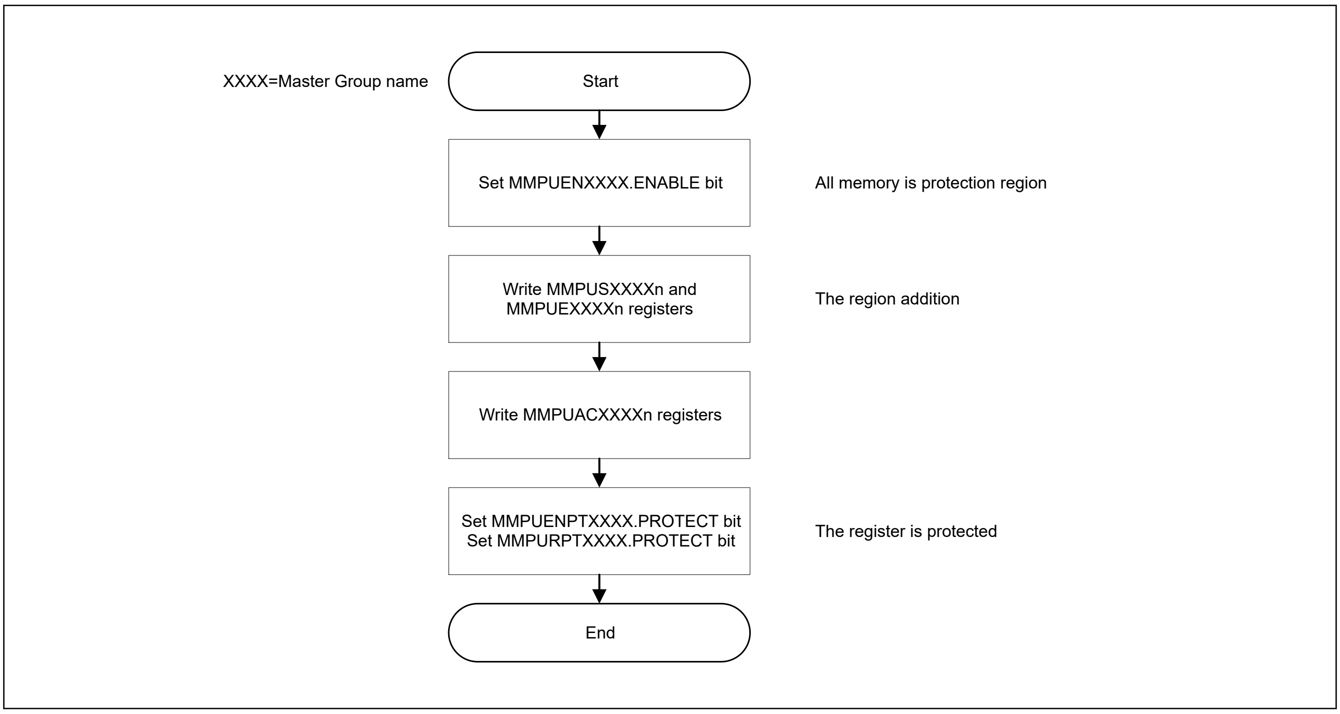

Figure 16.3 Register setting flow after reset

Figure 16.4 shows the register setting flow for adding regions. During this register setting, stop the master except the CPU.

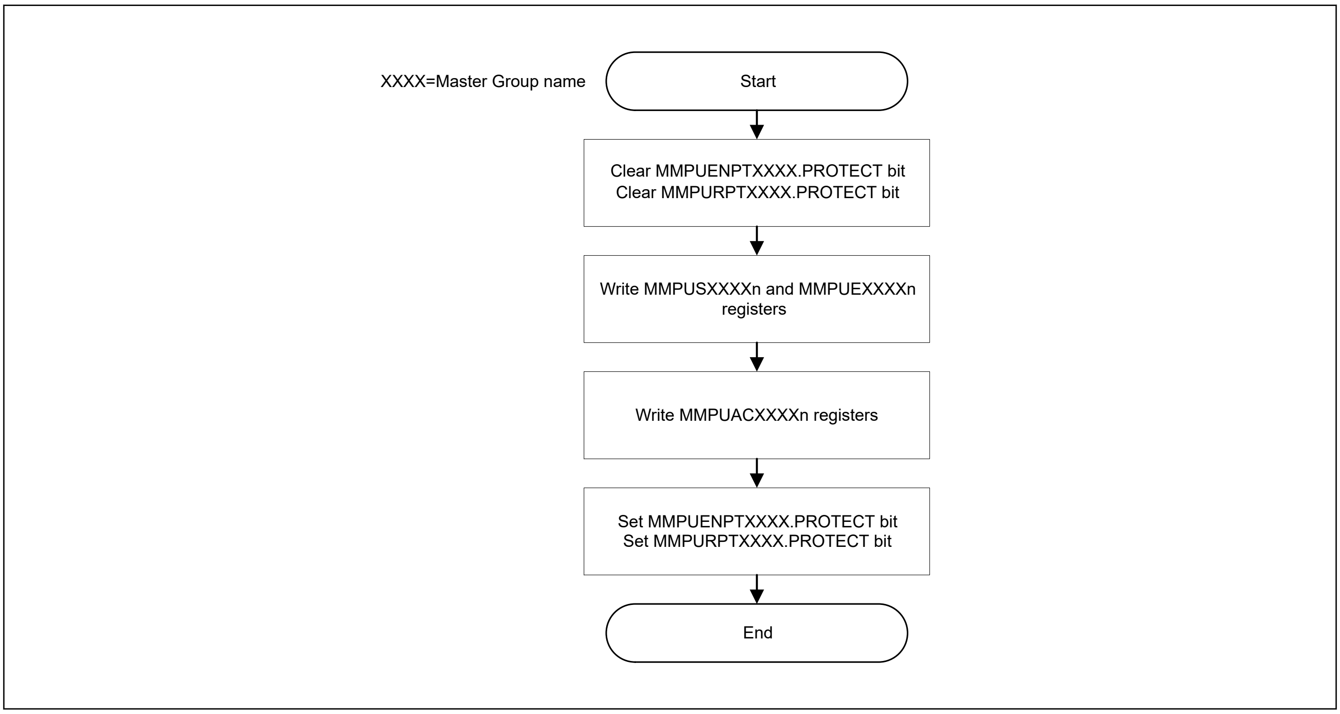

Figure 16.4 Register setting flow for region addition

16.3.2.2 Protecting the Registers

Registers related to the bus master MPU can be protected with the PROTECT bit as shown in Table 16.4.

Table 16.4 PROTECT bit and protect target registers (1 of 2)

PROTECT bit	Protect target registers
MMPUENPTDMACm.PROTECT	MMPUENDMACm (m = 0, 1)

Table 16.4 PROTECT bit and protect target registers (2 of 2)

PROTECT bit	Protect target registers
MMPUENPTEDMAC.PROTECT	MMPUENEDMAC
MMPUENPTGLCDC.PROTECT	MMPUENGLCDC
MMPUENPDRW.PROTECT	MMPUENDRW
MMPUENPTMIPID.PROTECT	MMPUENMIPID
MMPUENPTCEU.PROTECT	MMPUENCEU
MMPUENPTMIPIC.PROTECT	MMPUENMIPIC
MMPUENPTNPU.PROTECT	MMPUENNPU
MMPURPTDMACm.PROTECT	The following registers set to Non-secure by MMPUSARA.MMPUASAn (n = 0 to 7, 16 to 23). MMPUSDMACmk (m = 0, 1, k = 00 to 07) MMPUEDMACmk (m = 0, 1, k = 00 to 07) MMPUACDMACmk (m = 0, 1, k = 00 to 07)
MMPURPTDMACm_SEC.PROTECT	The following registers set to Secure by MMPUSARA.MMPUASAn (n = 0 to 7, 16 to 23). MMPUSDMACmk (m = 0, 1, k = 00 to 07) MMPUEDMACmk (m = 0, 1, k = 00 to 07) MMPUACDMACmk (m = 0, 1, k = 00 to 07)
MMPURPTEDMAC.PROTECT	MMPUSEDMACn (n = 0 to 4) MMPUEEDMACn (n = 0 to 4) MMPUACEDMACn (n = 0 to 4)
MMPURPTGLCDC.PROTECT	MMPUSGLCDCn (n = 0, 1) MMPUEGLCDCn (n = 0, 1) MMPUACGLCDCn (n = 0, 1)
MMPURPDRW.PROTECT	MMPUSDRWn (n = 0 to 2) MMPUEDRWn (n = 0 to 2) MMPUACDRWn (n = 0 to 2)
MMPURPTMIPID.PROTECT	MMPUSMIPID MMPUEMIPID MMPUACMIPID
MMPURPTCEU.PROTECT	MMPUSCEUn (n = 0, 1) MMPUECEUn (n = 0, 1) MMPUACCEUn (n = 0, 1)
MMPURPTMIPIC.PROTECT	MMPUSMIPICn (n = 0 to 2) MMPUEMIPICn (n = 0 to 2) MMPUACMIPICn (n = 0 to 2)
MMPURPTNPU.PROTECT	MMPUSNPUn (n = 0 to 4) MMPUENPUn (n = 0 to 4) MMPUACNPUn (n = 0 to 4)
MMPUOADPT.PROTECT	MMPUOAD.OAD

16.3.2.3 Memory Protection Error

If access to a protected region is detected, the bus master MPU generates an error. Set the MMPUOAD bit to select whether the error is reported as an IRQ or a reset.

For details of bus master MPU error, see section 15, Buses.

The IRQ status is indicated in ICU.IELSRn.IR. For details, see section 14, Interrupt Controller Unit (ICU). The reset status is indicated in SYSC.RSTSR1.BUSRF. For details, see section 6, Resets. Interrupts due to bus errors can also be used as NMI. For details of the specifications, see section 14.2.14. NMIER : Non-Maskable Interrupt Enable Register.

16.4 References

1. *ARM® v8-M Architecture Reference Manual* (ARM DDI0553B.g)
2. *ARM® Cortex®-M85 Processor Technical Reference Manual* (ARM 101924_0002_05_en)
3. *ARM® Cortex®-M85 Processor User Guide Reference Material* (ARM 101927_0002_05_en)

4. *ARM® Cortex®-M85 Processor Integration and Implementation Manual* (ARM 101925_0002_05_en)
5. *ARM® Cortex®-M33 Processor Technical Reference Manual* (100230_0004_00_en)
6. *ARM® Cortex®-M33 Processor User Guide Reference Material* (ARM 100234_0002_00_en)
7. *ARM® Cortex®-M33 Processor Integration and Implementation Manual* (ARM 100323_0002_00_en)